Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 1 / 方法 1 [PS01-M01]

Start by proving what is equally likely. Labeled samples, color patterns, multisets and stopped paths are different sample spaces; use the representation that makes the event easiest to count while preserving its weights.

首先证明什么结果等可能。有标签样本、颜色模式、多重集与停时路径属于不同样本空间；应选择最便于统计事件且保留权重的表示。

Choose equally likely labeled samples before counting colors

统计颜色前先选择等可能的有标签样本

Procedure

1. Decide whether the elementary outcome is an ordered draw, an unordered subset, a permutation, or a full deal.

2. For k draws without replacement from N distinct objects, use $\binom{N}{k}$ unordered outcomes or $N(N-1)\cdots(N-k+1)$ ordered outcomes consistently.

3. Count favorable group multiplicities with products of binomial coefficients; color words with different multiplicities need not have equal probability.

4. Divide favorable by total outcomes only after proving the chosen elementary outcomes are uniform.

通用步骤

1. 确定基本结果是有序抽取、无序子集、排列还是完整分配。

2. 从 N 个不同物品不放回取 k 个时，一致使用 $\binom{N}{k}$ 个无序结果，或 $N(N-1)\cdots(N-k+1)$ 个有序结果。

3. 用二项式系数之积统计有利类别重数；重数不同的颜色串未必等概率。

4. 证明所选基本结果均匀后，才能以有利数除以总数。

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Method 1 / 方法 1 [PS01-M01] (continued / 续)

> Example. An urn contains four red and three blue distinct balls. Find the chance that a uniformly chosen three-ball subset has exactly two red balls.

> 例题：袋中有四个红球和三个蓝球，球均可区分。均匀选择三球子集，恰有两红的概率是多少?

Worked solution

1. There are $\binom{7}{3} = 35$ equally likely subsets.

2. Choose two of the four red balls and one of the three blue balls: $\binom{4}{2}\binom{3}{1} = 18$.

3. The probability is $18/35$.

例题解析

1. 共有 $\binom{7}{3} = 35$ 个等可能子集。

2. 从四红中选二、三蓝中选一，有 $\binom{4}{2}\binom{3}{1} = 18$ 种。

3. 概率为 $18/35$。

Different color words can correspond to different numbers of labeled outcomes.

不同颜色串可能对应不同数量的有标签结果。

Practice / 配套练习: AIME-2000-II-Q03 (WB p. 720), AIME-2005-II-Q02 (WB p. 722), AIME-2010-I-Q01 (WB p. 723), AIME-2013-I-Q06 (WB p. 740), AIME-2015-I-Q02 (WB p. 726), AIME-2015-I-Q05 (WB p. 753), AIME-2018-I-Q03 (WB p. 728), AIME-2026-II-Q05 (WB p. 732)

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Method 2 / 方法 2 [PS01-M02]

Count discrete geometric pairs through edges, gaps or position sets

用边、间隔或位置集合统计离散几何配对

Procedure

1. Represent the sampled geometric objects by labeled vertices or position subsets.

2. Count admissible pairs by graph edges or by separation, preserving ordered versus unordered sampling.

3. For a cyclic configuration, identify a unique anchor or a family of disjoint opposite-position pairs to avoid overcounting.

4. Use exact endpoint conventions for open semicircles and strict triangle inequalities.

通用步骤

1. 用有标签顶点或位置子集表示被抽取的几何对象。

2. 按图的边或间隔统计允许配对，并保持有序或无序抽样一致。

3. 循环构型须确定唯一锚点，或使用互不相交的对位配对，避免重复计数。

4. 对开半圆与严格三角不等式使用准确端点约定。

> Example. A graph on vertices $1, 2, 3, 4$ has edges $12, 23, 31, 34$. Choose two distinct vertices uniformly. What is the probability they are joined by an edge?

> 例题：顶点为 $1, 2, 3, 4$ 的图具有边 $12, 23, 31, 34$。均匀选两个不同顶点，它们有边相连的概率是多少?

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 2 / 方法 2 [PS01-M02] (continued / 续)

Worked solution

1. The unordered sample space has $\binom{4}{2} = 6$ pairs.

2. The four listed edges are four distinct favorable pairs.

3. The probability is $4/6 = 2/3$. Ordered sampling doubles both counts and gives the same result.

例题解析

1. 无序样本空间有 $\binom{4}{2} = 6$ 对。

2. 所列四条边是四个不同的有利配对。

3. 概率为 $4/6 = 2/3$；有序抽样会同时使分子分母加倍，结果相同。

A discrete geometry question uses counts of choices; lengths alone do not weight the sample space.

离散几何题使用选择个数；长度本身不会给样本空间加权。

An obtuse inscribed triangle has its vertices in an open semicircle, with a unique first clockwise vertex. For an even polygon the antipodal endpoint is excluded, removing right triangles; for an odd polygon no vertex is exactly antipodal. Derive the two half-circle counts separately.

钝角内接三角形的顶点位于一个开半圆中，且有唯一顺时针首顶点。偶数边形须排除对径端点以去掉直角三角形；奇数边形没有恰好对径的顶点。应分别推导两种半圆计数。

Practice / 配套练习: AIME-2010-II-Q04 (WB p. 724), AIME-2011-I-Q10 (WB p. 739), AIME-2015-II-Q05 (WB p. 727), AIME-2019-I-Q02 (WB p. 729), AIME-2023-I-Q01 (WB p. 731)

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 3 / 方法 3 [PS01-M03]

Use product coordinates and inclusion-exclusion on event unions

在事件并集中使用乘积坐标与容斥

Procedure

1. Encode each independent elementary choice locally, such as membership bits or cell colors.

2. Count an intersection by the union of all forced conditions; shared objects are forced once, not once per event.

3. Apply inclusion-exclusion for overlapping good or bad events, and use a complement when simpler.

4. For a uniform random subset, every membership bit is independent with probability one half; conditioning on size changes that model.

通用步骤

1. 将每个独立基本选择作局部编码，例如成员位或格子颜色。

2. 交集须按全部被迫条件之并计数；共用对象只被固定一次，不能按事件重复固定。

3. 对重叠好事件或坏事件使用容斥；补集较简单时先数补集。

4. 均匀随机子集的成员位独立且各为二分之一；若附加大小条件，模型便改变。

> Example. Choose two independent uniform subsets A, B of a four-element set. Find $\Pr(|A \cap B| = 1)$.

> 例题：从四元素集合的所有子集中独立均匀选 A, B，求 $\Pr(|A \cap B| = 1)$。

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Method 3 / 方法 3 [PS01-M03] (continued / 续)

Worked solution

1. At each element the four membership pairs are equally likely, and only the pair $(1,1)$ enters the intersection.

2. The four elements act independently, so exactly one intersection element has probability $\binom{4}{1}(1/4)(3/4)^3$.

3. This simplifies to $27/64$.

例题解析

1. 每个元素的四种成员位对等可能，仅 $(1,1)$ 进入交集。

2. 四元素彼此独立，恰一个交集元素的概率为 $\binom{4}{1}(1/4)(3/4)^3$。

3. 化简为 $27/64$。

Inclusion-exclusion concerns overlapping events, even when the underlying random choices are independent.

即使基本随机选择独立，由它们构成的事件仍可能重叠而需要容斥。

Practice / 配套练习: AIME-2001-II-Q09 (WB p. 735), AIME-2002-I-Q01 (WB p. 721), AIME-2007-II-Q10 (WB p. 738), AIME-2011-II-Q12 (WB p. 745), AIME-2014-I-Q12 (WB p. 746), AIME-2025-II-Q07 (WB p. 492), AIME-2024-I-Q11 (WB p. 669)

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 4 / 方法 4 [PS01-M04]

Compress outcomes into weighted residue or exponent classes

将结果压缩为带权剩余类或指数类

Procedure

1. Choose a class label that exactly determines the event, such as parity, a prime-exponent vector or a residue sum.

2. Attach the correct weight to each class: the number or total probability of its elementary outcomes.

3. Count permitted class patterns, restoring multinomial ordering and class weights; remove repeated-object samples if distinctness is required.

4. For independent uniform exponent coordinates, count nondecreasing exponent tuples by combinations with repetition.

通用步骤

1. 选择能准确决定事件的类别标签，例如奇偶、质指数向量或剩余和。

2. 为每类附上正确权重，即其基本结果数量或总概率。

3. 统计允许类别模式，并恢复多项式排列权重与类权重；若要求对象不同，剔除重复抽到同一对象的样本。

4. 指数坐标独立均匀时，用可重复组合统计非降指数元组。

> Example. Two fair dice are rolled. Find the probability that their faces have the same parity but are unequal.

Section 5.1 Finite probability & combinatorial sampling

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Method 4 / 方法 4 [PS01-M04] (continued / 续)

> 例题：投两枚公平骰子，求点数奇偶相同但数值不同的概率。

Worked solution

1. There are 36 equally likely ordered outcomes. Each parity class has three faces.

2. Within either class, the first face has three choices and the distinct second face two choices, giving $2 \cdot 3 \cdot 2 = 12$ favorable outcomes.

3. The probability is $12/36 = 1/3$.

例题解析

1. 共有 36 个等可能有序结果，每个奇偶类有三个点数。

2. 每类首点数有三种，不同的第二点数有两种，有利结果为 $2 \cdot 3 \cdot 2 = 12$。

3. 概率为 $12/36 = 1/3$。

Compressing outcomes does not make the compressed classes equally likely.

压缩结果并不会使各压缩类别自动等概率。

For a square product, encode the parity of every prime exponent and require their coordinatewise sum to vanish modulo two. A face or number with zero parity vector may have more than one elementary representative and must retain that multiplicity.

For k equally likely exponent values, a nondecreasing r-tuple has $\binom{k+r-1}{r}$ possibilities out of k^r. Do not divide by $r!$, since tuples with repeated exponents have different orbit sizes.

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有限概率与组合抽样

Method 4 / 方法 4 [PS01-M04] (continued / 续)

乘积为平方数时，编码每个质指数的奇偶，并要求逐坐标和模二为零。零奇偶向量可能对应不止一个点数或数字，必须保留其重数。

指数有 k 个等可能值时，非降 r 元组有 $\binom{k+r-1}{r}$ 个，总数为 k^r。不能直接除以 $r!$，因为含重复指数的元组轨道大小不同。

Practice / 配套练习: AIME-2000-I-Q05 (WB p. 719), AIME-2001-I-Q06 (WB p. 733), AIME-2001-I-Q10 (WB p. 734), AIME-2004-II-Q10 (WB p. 736), AIME-2016-I-Q02 (WB p. 755), AIME-2018-I-Q12 (WB p. 748), AIME-2019-II-Q04 (WB p. 470), AIME-2020-I-Q09 (WB p. 741)

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Method 5 / 方法 5 [PS01-M05]

Count invariant colorings and independent orientation types

统计不变涂色与独立朝向类型

Procedure

1. For a fixed symmetry, partition positions into its permutation orbits.

2. An invariant coloring is constant on every orbit; if a color total is fixed, sum the orbit sizes rather than choose colors independently.

3. For many independent oriented objects, classify positions by how many outward constraints they impose and multiply the correctly weighted success probabilities.

4. For existence of some symmetry, count the union of successful symmetry families and resolve their intersections.

通用步骤

1. 对固定对称，将位置分成其置换轨道。

2. 不变涂色在每个轨道内恒定；若某颜色总数固定，须按轨道大小求和，不能独立选色。

3. 对许多独立定向物体，按位置施加的外露限制分类，再乘正确加权的成功概率。

4. 若要求存在某个对称，须数各成功对称族的并，并处理其交集。

> Example. Color the vertices of a square independently red or blue with equal probabilities. What is the probability of invariance under the reflection fixing one diagonal?

> 例题：独立以相等概率把正方形顶点涂红或蓝，求涂色在固定某条对角线的反射下不变的概率。

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Method 5 / 方法 5 [PS01-M05] (continued / 续)

Worked solution

1. The reflection fixes two vertices and exchanges the other two, giving orbit sizes $1, 1, 2$.

2. Each of three orbits can choose either color, so there are $2^3 = 8$ invariant colorings.

3. All $2^4 = 16$ vertex colorings are equally likely, so the probability is $1/2$.

例题解析

1. 该反射固定两个顶点，交换另两个，轨道大小为 $1, 1, 2$。

2. 三个轨道各可任选两色，故不变涂色有 $2^3 = 8$ 种。

3. 全部 $2^4 = 16$ 个顶点涂色等可能，故概率为 $1/2$。

Fixed color totals make the sample space a subset layer rather than independent bits.

固定颜色总数时，样本空间是固定大小子集层，而非独立成员位。

Practice / 配套练习: AIME-2005-I-Q09 (WB p. 737), AIME-2013-I-Q04 (WB p. 725), AIME-2024-I-Q11 (WB p. 669)

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有限概率与组合抽样

Method 6 / 方法 6 [PS01-M06]

Weight stopped paths by length and count constrained paths by state

按长度加权停时路径，并按状态统计受限路径

Procedure

1. Describe a terminal path including its stopping step; paths of different lengths generally have different probabilities.

2. For a fixed step multiplicity, use a multinomial count and multiply by that path's step probabilities.

3. For distinct-vertex or closing constraints, use a state recording the visited subset and last vertex, then test the closing edge.

4. Divide by a uniform total only when all retained elementary paths are equally likely.

通用步骤

1. 描述终止路径时包括停止步骤；不同长度路径通常具有不同概率。

2. 固定步型重数后，用多项式系数计数并乘该路径逐步概率。

3. 对顶点互异或闭合限制，使用记录已访问子集及末顶点的状态，再检验闭合边。

4. 只有保留的基本路径均等概率时，才能除以统一总数。

> Example. Toss a fair coin until the first head or three tosses, whichever occurs first. What is the probability that the terminal sequence contains more tails than heads?

> 例题：投公平硬币，直到首次正面或已投三次即停止。求终止串中反面多于正面的概率。

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 6 / 方法 6 [PS01-M06] (continued / 续)

Worked solution

1. The terminal paths are H, TH, TTH, TTT, with probabilities $1/2, 1/4, 1/8, 1/8$.

2. Only TTH and TTT have more tails than heads.

3. Add their probabilities to get $1/4$. Counting two favorable paths out of four would incorrectly treat unequal paths as uniform.

例题解析

1. 终止路径为 H, TH, TTH, TTT，概率分别为 $1/2, 1/4, 1/8, 1/8$。

2. 只有 TTH, TTT 的反面数多于正面数。

3. 相加得 $1/4$；按四条路径中两条有利计数会错误地把不等概率路径看成均匀。

The stopping rule is part of the sample-space definition.

停止规则本身就是样本空间定义的一部分。

Practice / 配套练习: AIME-2001-I-Q15 (WB p. 744), AIME-2019-I-Q05 (WB p. 730)

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 7 / 方法 7 [PS01-M07]

Model pairings and ownership constraints as permutations or matchings

把配对与归属限制建模为置换或匹配

Procedure

1. For a left-right bijection use a permutation; for arbitrary unordered pairings use a perfect matching with $(2n-1)!!$ outcomes.

2. Translate a closed collection of owners into a union of permutation cycles, or a sorted pair condition into a restriction on smaller endpoints.

3. Count allowable cycle types or pairing cases, correcting indistinguishable cycles and case intersections.

4. For repeated-color rows, use a uniform labeled model or prove every color pattern has the same number of labeled realizations.

通用步骤

1. 左右两类一一配对使用置换；任意无序配对使用完美匹配，共 $(2n-1)!!$ 种。

2. 将归属封闭子集转为置换循环之并，或将排序配对条件转为较小端点限制。

3. 统计允许循环型或配对情形，并修正不可区分循环与分类交集。

4. 对重复颜色行，使用均匀有标签模型，或证明每个颜色模式具有同数目的有标签实现。

> Example. Choose a uniform permutation of four labels. Find the probability that it has exactly one fixed point.

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Method 7 / 方法 7 [PS01-M07] (continued / 续)

例题：均匀选择四个标签的一个置换，求恰有一个不动点的概率。

Worked solution

1. Choose the fixed label in four ways.

2. The remaining three labels must have no fixed point, so they form one of the two three-cycles.

3. There are 8 favorable permutations out of $4! = 24$, giving probability $1/3$.

例题解析

1. 固定标签有四种选择。

2. 其余三个标签不能固定，因此构成两个三循环之一。

3. 有利置换为八个，全部为 $4! = 24$ 个，故概率为 $1/3$。

A pairing count and a permutation count use different elementary outcomes.

匹配计数与置换计数使用不同的基本结果。

Practice / 配套练习: AIME-2014-II-Q13 (WB p. 747), AIME-2022-I-Q09 (WB p. 742), AIME-2025-I-Q07 (WB p. 743)